

ROBERT GERVAIS

CURRICULUM VITAE

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OBJECTIVE

I led documentation for Unreal Engine, and I helped deliver the dream of desalination to fight drought. I am a programmer, engineer, and writer who delivers elegant solutions to complex problems, and I am looking to bring my toolbox to tackle something new.

TOOLBOX

WRITING	UNREAL ENGINE
Lua / C#	PROGRAMMING
ENGINEERING	C++ / Blueprints / Python

EDUCATION

Bachelor of Science in Computer Science
Real-time Interactive Simulations
DigiPen Institute of Technology (2016)

Bachelor of Science
Technology & Industrial Arts Management
Berea College (2002)

ADVISORY BOARDS

University of Washington
M.Sc. Information Management Program
Advising G.A.M.E.R. Group researchers on the development of the video game metadata schema, and sponsoring capstone projects at the iSchool.

foundry10
G.O.K.U. Metadata for Educational Games
Advising researchers on the development of a metadata schema that will enable libraries and educational institutions to categorize educational games.

CERTIFICATIONS

PSMJ Resources
Certified Project Management Professional
DPIC Companies

EXPERIENCE

Scientist

West Yost Associates

03/2025 – Present
Remote

- **Editing Technical Documents:** Providing technical editing for over 30 municipal agencies across the State of CA to develop Cross-Connection Control Plans which must meet requirements set by the State Department of Water Resources.
- **Providing Quality Control:** Performing quality control reviews of design drawings and specifications for cross-connection control, improvement, and fire service plans, including review of SCADA systems for automated pressure reducing station design.

Education & Learning Director

Epic Games

2023 – 03/2025
Remote

- **Championed Leadership:** Entrusted by executives to lead product documentation teams to unify all of the company's documentation under a single landing page.
- **Facilitated Collaboration:** I collaborated with user experience, engineering, design, art, and business teams to provide technical direction while building Epic's first AI-enabled chatbot (Epic Developer Assistant), which will integrate directly into Unreal Engine to bring the company's unified documentation into the product as a core learning feature.
- **Led Communication:** Primary point of contact for Disney, LEGO, Paramount, and more, to deliver documentation and learning resources in partnership with Epic's unified learning platforms.

Education & Learning Lead, Fortnite

Epic Games

2021 – 2023
Remote

- **Guaranteed 100% Feature Coverage:** Led efforts to work with Fortnite's engineers, artists, and designers to ensure full documentation coverage on launch day for Fortnite Creative, Unreal Editor for Fortnite, and the Verse programming language, utilizing TLA+ to validate algorithms written in a new multi-paradigm programming language featuring concurrency semantics. During the launch post-mortem, the UX team reported that a majority of users (>90%) shared that the documentation was functional and useful on launch day for all of Fortnite's debut UGC products.
- **Delivered Top Performing SEO:** Partnered with the SEO and UX teams to ship site indices, taxonomies, and glossaries that resulted in >90% CTR on the top 100 documentation pages, including over 80% engagement rates for pages in the lowest (viewed) quartile, indicating that the community was reliably finding documentation with organic search.
- **Designed Solution Architecture:** Designed the solution architecture for Direct Event Binding, eliminating the complexity of developing islands with the channels system, unlocking new game modes, and making it possible for creators to make new and entertaining Fortnite experiences faster than ever before.

Senior Technical Writer

Epic Games

2019 – 2021
Bellevue, WA

- **Implemented Information Architecture:** Ran open and closed card sort exercises, interviewed representatives from every team on Unreal Engine, utilized ANOVA to analyze company-wide card sort results, and ran A/B tests to design the information architecture for the launch of UE5.
- **Created Knowledge Graph:** Interviewed over 100 engineers, designers, and artists around the company to develop a comprehensive Simple Knowledge Organization System (SKOS) schema in Neo4j, which was leveraged to specify the Epic Developer Assistant.

Technical Writer

Epic Games

2017 – 2019
Bellevue, WA

- **Published White Paper:** Collaborated with the Lead Build Engineer to capture Epic's process of developing Fortnite, Paragon, and Battle Breakers, documenting key workflows associated with the development of UnrealGameSync, Epic's custom distributed development tool, which includes Perforce sync features, file conflict resolution, file versioning, project file generation, and optional binary builds and execution.

NOTABLE AWARDS

DigiPen Game Awards

Best Spoken Dialogue & Best Characters

Writing on Relic was voted by students in 2015 to receive 1st place for Best Spoken Dialogue and 1st place for Best Characters.

Zuda Finalist

DC Comics

My one-shot comic, entitled "Pirate Eye," was a finalist in the Zuda Comics competition held by DC Comics in 2009.

CELSOC & ACEC Awards

Engineering Excellence

The Encina Seawater Desalinated Conveyance Facilities Study was recognized for engineering excellence in 2006.

INDEPENDENT PROJECTS

Satoyama Dice

Lead Programmer

Programming a video poker simulator with Lua and writing OpenGL fragment shaders in LOVE for mini Sudoku puzzle games.

Demon Crush

Programmer & Narrative Designer

Programming gameplay systems in C++ and Blueprints, writing shaders in Custom HLSL, and designing levels in Tiled for a Paper2D pixel art beat-em up platformer game in Unreal Engine 5.

Seeds of Change

Writer

Published a 280-page post-apocalyptic graphic novel where plants are at the top of the food-chain.

OPEN SOURCE PROJECTS

ArticyXImporterForUnreal

Programming

Fixing bugs for ArticyXImporterForUnreal, an open source narrative design plugin.

ZILF

Documentation

Developing ontology and documentation for ZILF, which are tools for working with the ZIL interactive fiction language, including a compiler, assembler, disassembler, and game library.

STUDENT PROJECT

Relic

Narrative Lead & Programmer

Programmed UI in Zilch, ran writer's room, wrote all dialogue using structured XML files, directed voice over sessions, kept world building documentation, designed environments.

- **Created Style Guide:** While working on the white paper demonstrating the workflows and tools used to develop and publish Fortnite, I normalized Epic's voice, tone, and style to create the company's first style guide for written documentation, which was added to the original version of Epic's Brand Guidelines, setting the standard for how EULA users and licensees interface with Epic.

Training Content Developer

Epic Games

2015 – 2017

Bellevue, WA

- **Shipped Visual Studio Extension:** Developed a Visual Studio Extension that templated a Diataxis approach to writing documentation, which was leveraged to automatically generate the metadata, structured markdown syntax, and template language for improving the quality and quantity of documentation being generated for Unreal Engine.
- **Shipped Unreal Engine Plugin:** Developed an Unreal Engine plugin for our tutorial system, enabling engineers to wrap functional engine and gameplay code into snippets that APIDocTool could ingest for the C++ API Reference Manual.

Technical Writing Intern

Epic Games

Summer 2015

Cary, NC

- **Developed FPS Sample:** Mapped classes between Unreal Tournament and Unreal Engine into UML diagrams, which enabled me to migrate legacy components from Unreal Tournament into Unreal Engine's First Person Shooter sample, writing one of the most downloaded and widely used tutorials on Epic's documentation site.
- **Maintained APIDocTool:** Debugged Linux, Android, and iOS API calls into APIDocTool, building the hooks for Unreal's reflection system to detect new code being written for upcoming releases of UE4. I also maintained and extended the Code Snippet and Statistics modules as the engine grew over time.
- **Implemented Engineering Methods:** Implemented rigorous engineering methods, including lightweight RFC processes and data driven methods for developing valuable technical artifacts in order to standardize and elevate the quality of documentation being written for Unreal Engine 4.

"With the dream of desalination, delivered, I focused my skills to document Unreal Engine."

Engineer

Aegis Engineering Management

2009 – 2015

Remote

- **Desalination, Delivered:** With the successful bonding and construction of the largest desalination project in the Western Hemisphere, the Carlsbad "Bud" Lewis Desalination Facility was opened in December, 2015.
- **Expanded Recycled Water:** Wrote technical memorandums for municipal agencies to comply with Titles 17 & 22 of the CA Code of Regulations.
- **Validated Legal Descriptions:** Scripted tools in Python to validate plats and legal descriptions for public agencies, utilizing GIS databases to validate bearings reported by land surveyors.

Assistant Engineer II

AECOM

2008 – 2009

San Diego, CA

- **Supported Desalination:** Updated engineering documents, cost estimates, and technical artifacts to provide ongoing support for agencies addressing legal challenges and regulatory reviews that challenged the construction of the planned Carlsbad "Bud" Lewis Desalination Facility.
- **Estimated Costs:** Developed economic benefit apportionment reports for private and municipal agencies to comply with the CA Landscaping & Lighting Act of 1972 and CA Proposition 218.
- **Built Tools:** Engineered tools in Lisp to streamline and automate drafting tasks for designers to easily script workflows, which improved the quantity and quality of design documents.

Assistant Engineer I

Boyle Engineering Corporation

2003 – 2008

San Diego, CA

- **Planned Desalination:** On the heels of record drought conditions in Southern California, I was entrusted by senior leadership to manage 14 engineering firms to generate technical appendices for the Encina Seawater Desalinated Conveyance Facilities Study, which had an aggressive 6-month schedule from contract to completion.
- **Published Specifications:** Wrote technical specifications for municipal agencies to comply with requirements set by the CA Department of Water Resources and the EPA. I also wrote legal descriptions for grants and quitclaims in accordance with the ALTA Land Surveying requirements.
- **Programmed Models:** Programmed Bernoulli models in Fortran and ran custom hydraulic water system analyses to evaluate the configuration of proposed water systems and their suitability to serve proposed developments during fire flow and peak demand conditions.

"As a student, I told the story of Mammoth Cave crayfish, raising ecological awareness on the KY River."